

Three 2D strategies for anisotropic Laplace BEM

Status: ready-for-human

Orthotropic heat conduction on the unit square with a circular hole, compared to a FEniCS reference on the right edge $x = 1$. This note is **two-dimensional only**.

Package flux throughout: $q = -n \cdot K \nabla u$ (outward n). FEniCS stores $K \nabla T \cdot n$, so a right-edge influx of +5 is $q = -5$ in BEM.jl.

The problem

$$\nabla \cdot (K \nabla u) = 0 \quad \text{in } \Omega, \quad K = \begin{pmatrix} 5 & 0 \\ 0 & 0.5 \end{pmatrix}.$$

Ω is the unit square minus a disk of radius 0.25 at (0.5, 0.5). Mesh builder: `placa_furo_orto`. Quadratic discontinuous collocation (`ordem=2`, `tipo=2`).

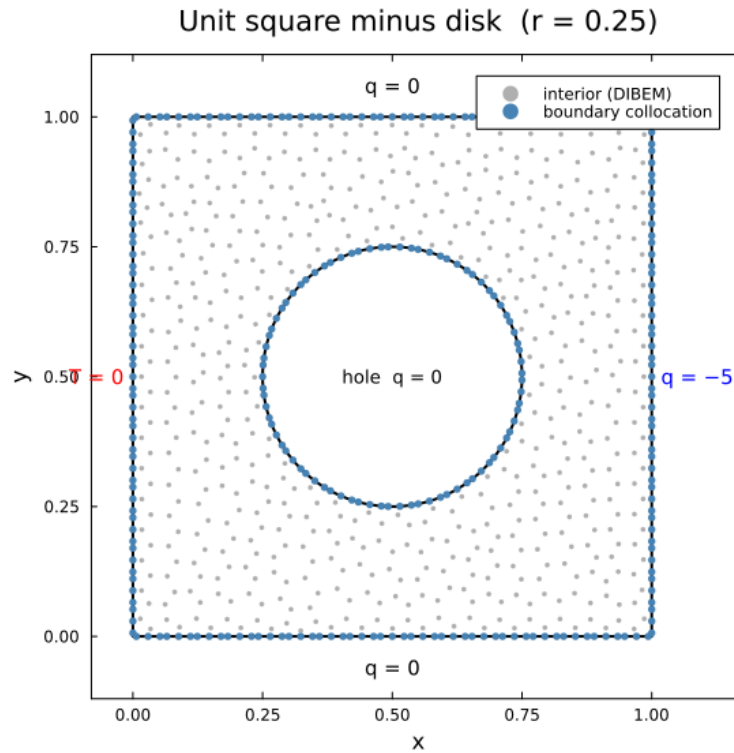


Figure 1: Domain and BCs. Red: Dirichlet $T = 0$ on $x = 0$. Blue: Neumann $q = -5$ on $x = 1$ (FEniCS $K \nabla T \cdot n = 5$). All other edges, including the hole, are insulated ($q = 0$). Grey points are interior collocation used by DIBEM (S1 and S3). S2 uses the boundary only.

The hole constricts the flux, so $T(x = 1, y)$ peaks near $y = 0.5$. FEniCS ($h = 0.02$) gives $T_{\text{mid}} = T(1, 0.5) = 2.769$. Without the hole one would have $T = x$ and $T(1, y) = 1$.

Reference file: `data/Laplace/fenics_ortho_plate_Tright.csv`.

Isotropic collocation BEM (shared by S1 and S3)

For isotropic conductivity k the collocation BIE is

$$c(\xi)u(\xi) + \int_{\Gamma} u \frac{\partial \Phi}{\partial n} d\Gamma = \int_{\Gamma} q_{\text{iso}} \Phi d\Gamma + \text{volume terms},$$

with $\Phi = -\ln \frac{r}{2\pi k}$ and package $q_{\text{iso}} = -k \frac{\partial u}{\partial n}$. In matrix form, after moving the free term into the diagonal of H ,

$$Hu - Gq_{\text{iso}} = \text{rhs}.$$

S1 and S3 assemble this system with `Laplace(1)` (so $k = 1$ in Φ), then **correct** q_{iso} and the volume residual for anisotropy. S2 never uses this isotropic pair: it replaces Φ by the anisotropic Green's function.

DIBEM supplies a mass matrix M such that, for a domain source f ,

$$Hu - Gq_{\text{iso}} = Mf,$$

with f interpolated by PHS radial basis functions and the volume integrals taken analytically in the radial coordinate (`int(rbf,x,xj)`) and the RIM primitive of Φ).

Strategy 2 — anisotropic fundamental solution

Direct collocation with the anisotropic Green's function (`AnisotropicLaplace(K)`). No DIBEM, no interiors, no Neumann map.

$$\Phi(r) = -\frac{\ln \sqrt{r \cdot K^{-1}r}}{2\pi \sqrt{\det K}}, \quad \rho^2 = r \cdot K^{-1}r.$$

The dual kernel is the **physical** flux of Φ :

$$q^* = -n \cdot K \nabla \Phi.$$

Because $KK^{-1}r = r$,

$$\nabla \Phi = -\frac{K^{-1}r}{2\pi \sqrt{\det K} \rho^2}, \quad q^* = \frac{r \cdot n}{2\pi \sqrt{\det K} \rho^2}.$$

The numerator is $r \cdot n$, not $n \cdot K^{-1}r$. The isotropic limit $K = kI$ recovers `Laplace(k)` (H has no extra k in the denominator, as in `Fundamental.jl`).

The linear system is the usual mixed BEM: $Hu = Gq$ with this (Φ, q^*) , Dirichlet columns swapped for $-G$.

Strategy 1 — isotropic Poisson + DIBEM residual (Hessian)

Keep the **isotropic** kernels and push anisotropy into a domain source. With $K = kI + \Delta K$,

$$\nabla \cdot (K \nabla u) = k \nabla^2 u + \nabla \cdot (\Delta K \nabla u) = 0,$$

so

$$\nabla^2 u = -f^*, \quad f^* = \left(\frac{1}{k}\right) \nabla \cdot (\Delta K \nabla u) \approx A_f u.$$

Package DIBEM identity $Hu - Gq_{\text{iso}} = M \nabla^2 u$ then becomes

$$Hu - Gq_{\text{iso}} = -M A_f u \quad (b = 0),$$

or $(H + M A_f)u - Gq_{\text{iso}} = 0$. The Neumann map replaces q_{iso} as above.

A_f is built from a Hessian of nodal u . A **global** PHS interpolant through the hole gives $|f^*| \sim O(100)$ even when the true residual is modest. S1 therefore uses **local** RBF-FD (`nlocal = 21`, PHS3 with `poly_deg=2`) on all collocation points (boundary + cell centroids). The Hessian of the

polynomial tail is filled ($x^2 \rightarrow 2, xy \rightarrow 1, \dots$). On the coarsest mesh ($n = 168$) that stencil is too thin around the hole and S1 fails; medium and fine meshes are fine.

S1 **does** form $\text{Hess}(u)$. That is the difference from S3.

Strategy 3 — one integration by parts, then DIBEM (no Hessian)

Do **not** differentiate u twice. Integrate $\nabla \Delta K \nabla u$ by parts once against the isotropic fundamental solution Φ :

$$\int_{\Omega} \Phi f^* d\Omega = \left(\frac{1}{k}\right) \int_{\Gamma} \Phi (n \cdot \Delta K \nabla u) d\Gamma - \left(\frac{1}{k}\right) \int_{\Omega} \nabla \Phi \cdot (\Delta K \nabla u) d\Omega.$$

Three discrete pieces, all existing DIBEM machinery:

1. $\int_{\Omega} \Phi b$: the mass M (here $b = 0$).
2. $\int_{\Gamma} \Phi (n \cdot \Delta K \nabla u)$: G times the same split $n \cdot \Delta K \nabla u = S_{\Gamma} u - n \Delta n q_{\text{iso}}$ as in the Neumann map.
3. $\int_{\Omega} \nabla \Phi \cdot w$ with $w = \Delta K \nabla u$: Loeffler operator N_{α} whose kernel is $\partial_{\alpha} \Phi$. The radial integrals use the same $\text{int}(\text{rbf})$ coefficients c as M , and the RIM primitive of $\nabla \Phi$ is $-R \frac{e}{2\pi k}$ in 2D ($e = \frac{r}{R}$).

∇u is an RBF interpolant of **values** (first derivatives only). There is no Hessian of u .

After collecting terms,

$$Hu - Gq_{\text{iso}} + \left(\frac{1}{k}\right) G(S_{\Gamma} u - n \Delta n q_{\text{iso}}) - \left(\frac{1}{k}\right) A_V u = 0,$$

with $A_V = \sum_{\alpha, \beta} \Delta K_{\alpha\beta} N_{\alpha} D_{\beta}$ and $D_{\beta} u \approx \partial_{\beta} u$. The Neumann map is then applied exactly as in S1.

Right-edge profiles vs FEniCS

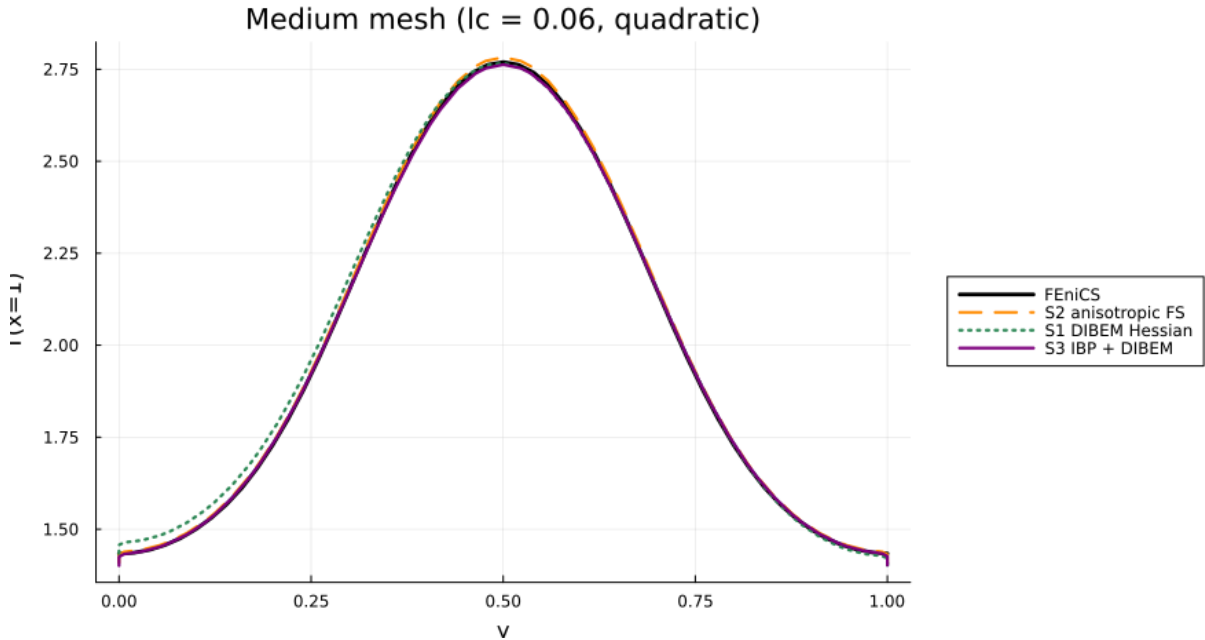


Figure 2: Medium mesh ($lc = 0.06, n = 285$ quadratic). All three strategies follow the FEniCS bump. S1 is slightly high on the lower ligament ($y < 0.4$); S2 and S3 lie on top of FEniCS.

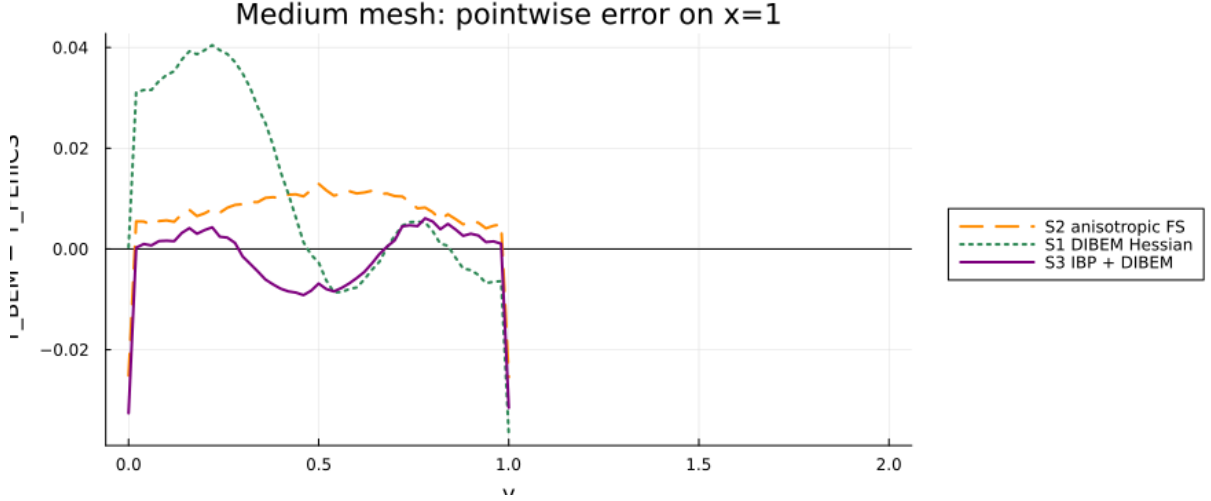


Figure 3: Pointwise error $T_{\text{BEM}} - T_{\text{FEniCS}}$ on $x = 1$, same mesh. S2 error is a small positive bias (~ 0.005 – 0.01). S3 stays within ± 0.01 except at the corners. S1 reaches $+0.04$ on the lower half.

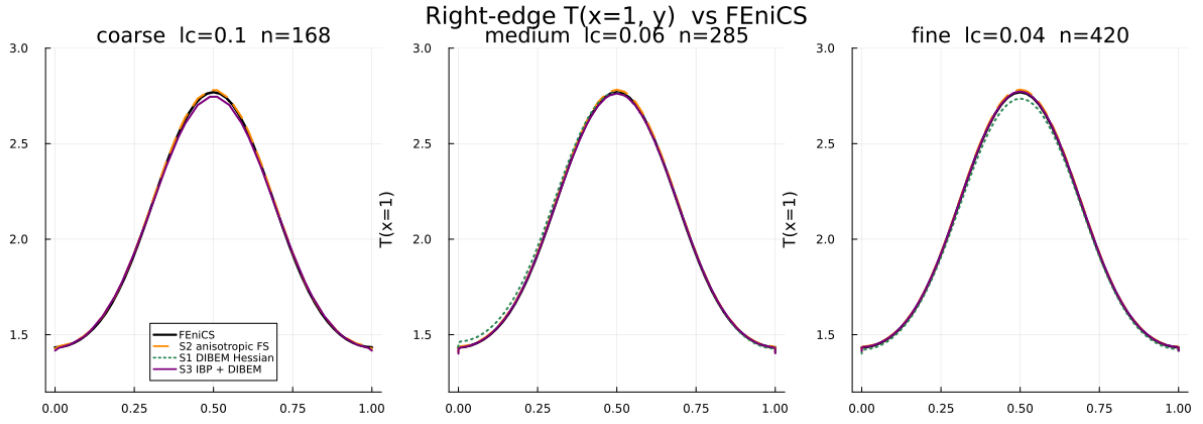


Figure 4: Three meshes. S2 and S3 are stable from coarse to fine. Coarse S1 is **not** plotted on scale: $T_{\text{mid}} = -1.57$ (RMS 1.23) because the local Hessian stencil is too small around the hole. Medium and fine S1 recover the bump (RMS ≈ 0.011).

Table (FEniCS $T_{\text{mid}} = 2.769$)

mesh	lc	n	S2 RMS / T_{mid}	S1 RMS / T_{mid}	S3 RMS / T_{mid}
coarse	0.10	168	0.0040 / 2.780	1.23 / -1.57	0.0070 / 2.745
medium	0.06	285	0.0048 / 2.782	0.0108 / 2.767	0.0038 / 2.763
fine	0.04	420	0.0044 / 2.782	0.0104 / 2.736	0.0025 / 2.773

Table 1: Relative L^2 error of $T(x = 1, y)$ vs FEniCS, and the value at $y = 0.5$.

S2 is **not** larger than FEniCS on these 2D models: T_{mid} is 2.78 against 2.77 (relative 0.5%). S3 is the closest of the three on medium and fine meshes. S1 needs a mesh fine enough for the local Hessian; once that holds, it is within about 1% RMS.

Manufactured anisotropic Poisson (known solution)

The holed-plate comparison has no closed-form u . To measure **convergence** of the two DIBEM strategies, use the manufactured field on the unit square (no hole)

$$u = \sin(\pi x) \sin(\pi y), \quad K = \begin{pmatrix} 5 & 0 \\ 0 & 0.5 \end{pmatrix}.$$

Then $\nabla u = \pi(\cos(\pi x) \sin(\pi y), \sin(\pi x) \cos(\pi y))$ and

$$\nabla \cdot (K \nabla u) = -\pi^2(k_x + k_y)u,$$

so the package source in $\nabla \cdot (K \nabla u) = -b$ is $b = \pi^2(k_x + k_y)u$. Physical flux $q = -n \cdot K \nabla u$. All-Dirichlet data from u ; the error is the relative L^2 norm on **interior** collocation (boundary u is prescribed). Quadratic elements (ordem=2, tipo=2). Helper: ana_aniso_poisson_sin.

S2 is omitted: the anisotropic Green's function treats the homogeneous BIE, so a volume source has to go through DIBEM (S1 or S3).

ndiv	h	n	n_i	S1 error	S3 error
4	0.333	36	9	1.57×10^{-1}	1.28×10^{-1}
6	0.200	60	25	4.82×10^{-2}	3.29×10^{-2}
8	0.143	84	49	1.11×10^{-2}	1.25×10^{-2}
12	0.091	132	121	5.67×10^{-3}	4.04×10^{-3}
16	0.067	180	225	3.15×10^{-3}	2.03×10^{-3}

Table 2: Interior relative L^2 error. $h = 1/(\text{ndiv} - 1)$.

Observed order from the last two meshes: S1 ≈ 1.9 , S3 ≈ 2.2 (both track $O(h^2)$). S3 is slightly more accurate on the finer meshes.

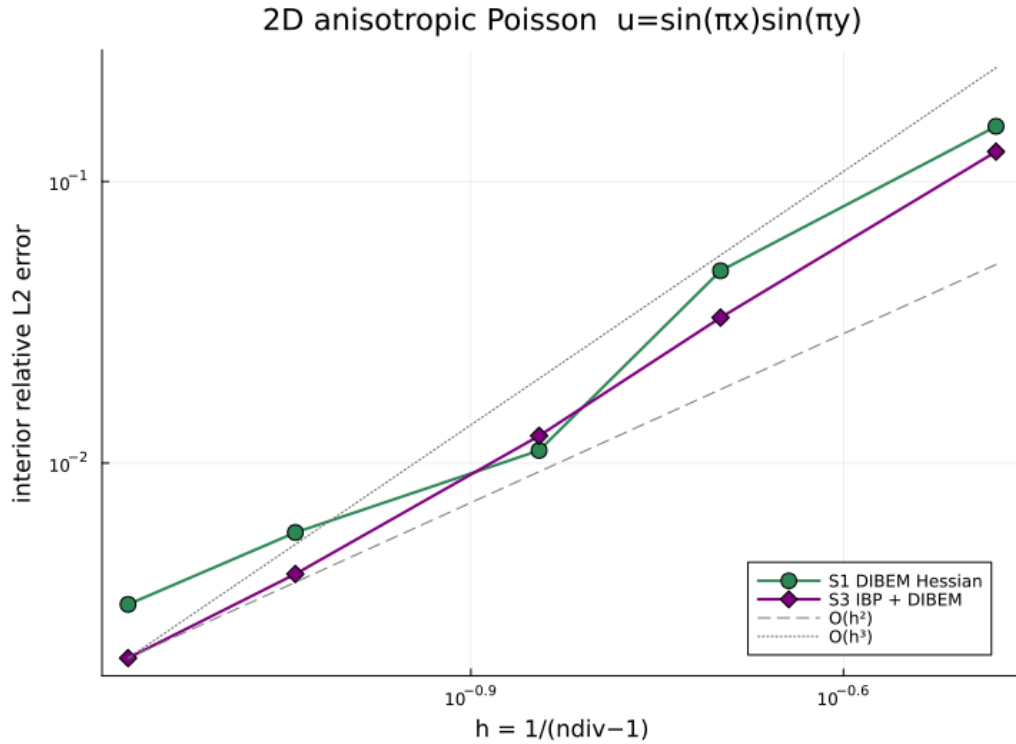


Figure 5: Log-log interior error versus h . Green: S1 (Hessian residual). Purple: S3 (IBP, no Hessian). Dashed $O(h^2)$, dotted $O(h^3)$.